

Lens-System Diffraction Integral Written in Terms of Matrix Optics*

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This paper deals with the propagation of coherent light through lens systems; it relates diffraction theory to ray optics. A diffraction integral is derived which relates the electromagnetic fields on the input plane of a lens system to those on its output plane. The kernel of the diffraction integral is written in terms of the elements of the ray matrix that describes the complete lens system; that kernel indicates a connection between ray optics and diffraction theory. It also provides a simple method for writing the diffraction integral for a lens system. The results are limited to the paraxial-ray approximation, but apply to symmetric and asymmetric lens systems. In the case of asymmetric systems, i.e., those containing rotated elliptical or cylindrical lenses, the ray-matrix formalism is extended so as to use a single fourth-order matrix. The diffraction integrals derived are applied to optical spatial filtering, optical-beam waveguides, optical resonators, and holography.

INDEX HEADINGS: Lens design; Diffraction; Ray tracing; Filter; Holography.

Lens systems can be analyzed by use of either ray optics or diffraction theory; the choice of the approach depends on the particular situation. A connection between ray optics and diffraction theory could facilitate calculations by allowing use of the simpler techniques of each approach. This paper provides such a connection and extends the two to cases of reduced symmetry.

Examples of the use of diffraction theory in lens systems occur in the calculation of electromagnetic fields in optical resonators¹ and in optical data-processing systems.² In each of these cases, a diffraction integral relates the electromagnetic fields on the input plane of an optical system to those on its output plane. In this paper, we generalize such integral relationships, presenting a diffraction integral for an arbitrary optical system, limited by the paraxial-ray restriction.

The associated ray theory is couched in terms of ray-matrix optics,^{3,4} in which refraction and translation matrices are multiplied together to form a complete-optical-system ray matrix. The ray-matrix formalism provides the generality necessary because it allows the diffraction-integral kernel to be written in terms of the complete-optical-system ray matrix.

This analysis is sufficiently general to extend ray-matrix optics to cases of systems with reduced symmetry, such as systems with tipped elliptical lenses. This will be especially useful in analysis of more general types of beam waveguides and in holography.

Diffraction theory is necessary in the discussion of lens systems for which phase information is required in addition to amplitude information. With the development of the laser, it is now possible to measure the relative phase of light as well as its amplitude. In areas

of study such as holography and optical resonators, it is necessary to consider phase terms in calculations. In other areas of study, such as optical spatial filtering, it is of interest to have analytical expressions for the phase of the associated electromagnetic fields.

The analytical techniques for the prediction of phase behavior of lens systems have received less consideration than have amplitude-prediction techniques because they have not until recently been required, the amplitude information provided by ray descriptions usually sufficing. To effect a complete physical description, including phase, it is necessary to use diffraction theory, relating fields at input and output planes by a diffraction integral. The integral presented herein gives the desired relationship in general terms, applicable to the variety of lens systems indicated.

This formalism presents quite a simple way of examining such diffraction problems involving lens systems. It is merely necessary to multiply the relevant ray matrices together to find the complete-system matrix, and to employ the integrals presented.

Previous work on coherent-light propagation through lens systems has been considered in studies of optical resonators^{1,5,6} and optical transmission lines.⁷⁻¹¹ These studies have generally been restricted to gaussian beams. The connection between the diffraction-theory description of lens systems and the ray-matrix description has been indicated by Brouwer *et al.*¹² Their results are, in principle, not limited to paraxial rays, but they consider only symmetric systems, and do not include all of the phase terms or the amplitude terms. Suematsu and Fukinuki¹³ have considered asymmetric optical systems; their work applies to gaussian-beam propagation. The results presented here are comparable with those of Walther.¹⁴ However, his omission of direct reference to ray-matrix optics impedes the application developed in this paper. Baus¹⁵ has formulated the optical-resonator integral equation in terms of ray-matrix equations, but only for systems with two perpendicular planes of symmetry. Vander Lugt¹⁶ has worked out a formal scheme for analyzing optical-system fields. His aim is to examine focusing and Fourier-transform relationships of optical systems; he made no attempt to relate his results to ray optics.

The present paper combines the diffraction theory necessary to describe optical resonators and optical transmission lines with the theory of Brouwer, which makes the connection with ray matrices. Both are extended to the case of asymmetric optical systems. The general integral relationship is derived and several examples are given, which predict and extend previous results.

To illustrate the type of results presented, the following expression, derived as Eq. (28), is quoted. If the geometrical-optics ray-matrix relationship between input and output planes of a symmetric optical

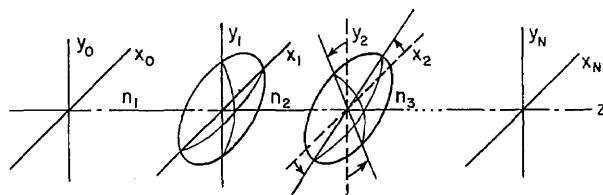


FIG. 1. Schematic illustration of coordinates, input plane x_0, y_0 , output plane x_N, y_N , and lens coordinates. The right-hand lens is rotated about the optic axis, to represent the conditions of reduced symmetry.

system is

$$\begin{pmatrix} n_N p_N \\ x_N \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} n_1 p_1 \\ x_0 \end{pmatrix},$$

then the diffraction integral relating the fields across these two planes is

$$E_N(x_N, y_N) = -\frac{ik}{2\pi C} \exp(ikL_0) \iint E(x_0, y_0) \times \exp \left\{ \frac{ik}{2C} [D(x_0^2 + y_0^2) - 2(x_0 x_N + y_0 y_N) + A(x_N^2 + y_N^2)] \right\} dx_0 dy_0,$$

where L_0 is the axial optical distance between the two planes.

I. PHYSICAL SITUATION AND DESCRIPTION OF APPROACH

The physical situation is shown in Fig. 1. Let $E_0(x_0, y_0)$ be the field on the input plane denoted by x_0, y_0 located at z_0 . This field propagates through the optical system to the output plane, x_N, y_N , located at z_N . The optical system is represented by smooth dielectric air interfaces, which define lens surfaces that intersect the optical axis at $z_{01}, z_{02}, \dots, z_{0N-1}$. The axes $x_0, x_1, \dots, x_N$ and $y_0, y_1, \dots, y_N$ are, respectively, parallel. The surfaces at $z_{01}, z_{02}, \dots, z_{0N-1}$ are not necessarily symmetric with respect to the x and y directions. The object is to find an integral relationship between $E_0(x_0, y_0)$ and the field on the output plane, $E_N(x_N, y_N)$.

It is assumed that there are no limiting apertures. This restriction can be removed by finding the field at the plane of the limiting aperture, by use of the integral derived, multiplying by the aperture function, and applying the integral a second time to describe propagation through the remainder of the system.

The first step in finding the desired integral is to note that it can be expressed in the form

$$E_N(x_N, y_N) = \iint E_0(x_0, y_0) K(x_0, y_0; x_N, y_N) dx_0 dy_0. \quad (1)$$

This follows from the linearity of Maxwell's equations. The kernel $K(x_0, y_0; x_N, y_N)$ is the point-response function, that is, the field on the output plane due to a point source at x_0, y_0 . Our job simplifies to finding the form of the point-response function.

To find $K(x_0, y_0; x_N, y_N)$, we assume a point source at x_0, y_0 and follow the mathematical formalism used in the development of the eikonal equation,¹⁷ assuming a transverse scalar field. Thus, in any region $z_i', z_{0i-1} \leq z_i' \leq z_{0i}$, the field, E_i , due to the point source is represented in the form

$$E_i = A_i \exp(ikL_i). \quad (2)$$

The amplitude function, A_i , and the phase or eikonal function, L_i , are real functions of position and $2\pi/k = \lambda_c$ is the free-space wavelength. A_i and L_i satisfy the following equations,¹⁸ derived from the wave equation:

$$|\text{grad} L_i| = n_i \quad (3)$$

$$2 \text{ grad} A_i \cdot \text{grad} L_i - \nabla^2 A_i = 0; \quad (4)$$

$K(x_0, y_0; x_N, y_N)$ is then $E_N(x_N, y_N)$ evaluated at the output plane, $z_N' = z_{0N}$. To find K , it is necessary to find A_N and L_N on the output plane, due to the point source at x_0, y_0 .

The approach to finding K is equivalent to solving the wave equation to obtain solutions in each region and then satisfying the appropriate boundary conditions. The boundary conditions to be satisfied at the interfaces are that the tangential component of E and the normal component of D be continuous. Only the first condition is significant, however, because, with the approximation considered, there is no significant normal component of D . The boundary condition on E then requires that L and A be continuous at the interfaces.

Taking the phase at the point source to be zero, we recognize that L_N is the optical distance along the ray connecting the source point and the point x_N, y_N in the output plane.¹⁹ Instead of solving for L_i in each region, we use the fact that the regions are homogeneous, so that the rays are straight lines. Fermat's principle is then used to choose the proper intersections of the ray path with the interfaces. This procedure is equivalent to applying the boundary conditions to the electromagnetic phase at the interfaces. The resulting optimized distance is the desired expression for L_N . This solution for L_N is formally given in Sec. II. Instead of carrying the solution through to completion, as described, a connection is made with ray-matrix optics and the solution for L_N is completed in terms of ray-matrix optics. This is accomplished in Sec. III.

A_N is obtained by using the conservation of energy to relate the fields on the input and output planes. This is equivalent to requiring that A be continuous at each interface because the requirement on the continuity of A is itself equivalent to conservation of energy law. This is done in Sec. IV. Section V contains some applications of the results.

II. DERIVATION OF EIKONAL FUNCTION, L_N

In order to use Fermat's principle, assuming straight-line ray propagation in each region, we consider a ray connecting the source point, with x_N, y_N and intersecting the i th surface at x_i, y_i, z_i . If we assume $(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2 \ll (z_i - z_{i-1})^2$, then the optical distance from the source at x_0, y_0 to x_N, y_N is

$$L_N' = \sum_{i=1}^N n_i \left(z_i - z_{i-1} + \frac{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2}{2(z_i - z_{i-1})} \right). \quad (5)$$

Let the surface at z_{0i} have the form

$$z_i = z_{0i} + c_i x_i^2 + d_i x_i y_i + e_i y_i^2. \quad (6)$$

The process of optimizing L_N' with respect to the x_i and y_i is considered in Appendix A. It results in the following set of equations for the optimized values of x_i and y_i , call them x_{0i} and y_{0i} , respectively.

$$\begin{aligned} \frac{\partial L_N'}{\partial x_i} = 0 = & -\frac{n_i x_{0i-1}}{(z_{0i} - z_{0i-1})} + x_{0i} \left[\frac{n_i}{(z_{0i} - z_{0i-1})} \right. \\ & \left. + \frac{n_{i+1}}{(z_{0i+1} - z_{0i})} - 2c_i(n_{i+1} - n_i) \right] \\ & - \frac{n_{i+1} x_{0i+1}}{(z_{0i+1} - z_{0i})} - y_{0i} d_i(n_{i+1} - n_i) \end{aligned} \quad (7a)$$

$$\begin{aligned} \frac{\partial L_N'}{\partial y_i} = 0 = & -\frac{n_i y_{0i-1}}{(z_{0i} - z_{0i-1})} + y_{0i} \left[\frac{n_i}{(z_{0i} - z_{0i-1})} \right. \\ & \left. + \frac{n_{i+1}}{(z_{0i+1} - z_{0i})} - 2e_i(n_{i+1} - n_i) \right] \\ & - \frac{n_{i+1} y_{0i+1}}{(z_{0i+1} - z_{0i})} - x_{0i} d_i(n_{i+1} - n_i). \end{aligned} \quad (7b)$$

Let the optimized expression for L_N' be denoted by

$$\begin{aligned} L_N = \sum_{i=1}^N n_i (z_{0i} - z_{0i-1}) & + \frac{n_1 x_0 (x_0 - x_{01})}{2(z_{01} - z_{00})} + \frac{n_1 y_0 (y_0 - y_{01})}{2(z_{01} - z_{00})} \\ & + \frac{n_N x_N (x_N - x_{0N-1})}{2(z_{0N} - z_{0N-1})} + \frac{n_N y_N (y_N - y_{0N-1})}{2(z_{0N} - z_{0N-1})}. \end{aligned} \quad (8)$$

Equation (8) is not quite in final form. It still contains the unknown quantities x_{01}, y_{01}, x_{0N-1} , and y_{0N-1} . However, enough information is now available to determine the minimum value, L_N . We can solve for x_{01}, y_{01}, x_{0N-1} , and y_{0N-1} in terms of x_0, y_0, x_N, y_N and the constants of the optical system using Eqs. (7) and substitute into Eq. (8). In the next section, the solutions will actually be obtained, but a method that makes the connection with ray-matrix theory will be used.

III. CONNECTION WITH MATRIX RAY OPTICS

To indicate the connection with ray-matrix optics, we start by deriving the ray-matrix equations for asymmetric lens systems. To do this, define slopes p_i and q_i ,

$$p_i = (x_{0i} - x_{0i-1}) / (z_{0i} - z_{0i-1}) \quad (9a)$$

$$q_i = (y_{0i} - y_{0i-1}) / (z_{0i} - z_{0i-1}). \quad (9b)$$

Then, in terms of p_i , q_i , p_N , and q_N , the expression for L_N in Eq. (8) reduces to

$$L_N = L_0 + \frac{1}{2}(-n_1 p_1 x_0 - n_1 q_1 y_0 + n_N p_N x_N + n_N q_N y_N), \quad (10)$$

where

$$L_0 = \sum_{i=1}^N n_i (z_{0i} - z_{0i-1}). \quad (11)$$

L_0 is the axial optical distance. Equations (7) can be rewritten as

$$n_i p_i - n_{i+1} p_{i+1} - 2c_i x_{0i} (n_{i+1} - n_i) - d_i y_{0i} (n_{i+1} - n_i) = 0 \quad (12a)$$

$$n_i q_i - n_{i+1} q_{i+1} - 2e_i y_{0i} (n_{i+1} - n_i) - d_i x_{0i} (n_{i+1} - n_i) = 0. \quad (12b)$$

Equations (12) define the change of ray slope from p_i in region z_i' to p_{i+1} in region z_{i+1}' as the ray passes through the refraction surface z_{0i} . We use the fact that the ray position does not change in passing through the interface at z_{0i} . This is formally stated in Eqs. (13), where x_{0i}' , and x_{0i}'' are, respectively, the ray positions immediately to the left and right of the surface at z_i ,

$$x_{0i}' = x_{0i}'' \quad (13a)$$

$$y_{0i}' = y_{0i}'' \quad (13b)$$

Equations (12) and (13) are then used to define the ray-matrix equation for the refracting surface at z_i ,

$$\begin{pmatrix} n_{i+1} p_{i+1} \\ x_{0i}'' \\ n_{i+1} q_{i+1} \\ y_{0i}'' \end{pmatrix} = \begin{pmatrix} 1 & -2c_i(n_{i+1} - n_i) & 0 & -d_i(n_{i+1} - n_i) \\ 0 & 1 & 0 & 0 \\ 0 & -d_i(n_{i+1} - n_i) & 1 & -2e_i(n_{i+1} - n_i) \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} n_i p_i \\ x_{0i}' \\ n_i q_i \\ y_{0i}' \end{pmatrix}. \quad (14)$$

Equation (14) is necessarily a 4×4 matrix transformation and not two independent 2×2 matrix transformations for the x and y directions, as usually presented,^{3,4} because of the $-d_i(n_{i+1} - n_i)$ terms that couple the two transverse directions. The ray matrix for a translation between refracting surfaces is, however, the same as that

for the two directions taken independently,

$$\begin{pmatrix} n_{i+1} p_{i+1} \\ x_{0i+1} \\ n_{i+1} q_{i+1} \\ y_{0i+1} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ \frac{(z_{0i+1} - z_{0i})}{n_i} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{(z_{0i+1} - z_{0i})}{n_i} & 1 \end{pmatrix} \begin{pmatrix} n_i p_i \\ x_{0i} \\ n_i q_i \\ y_{0i} \end{pmatrix} \quad (15)$$

Thus, following standard ray-matrix techniques, the translation and refraction matrices are multiplied together to give the system matrix relating slopes and positions at the input and output planes. Let such a product matrix for a complete optical system be called **T**; then

$$\begin{pmatrix} n_N p_N \\ x_N \\ n_N q_N \\ y_N \end{pmatrix} = \begin{pmatrix} A_x & B_x & E_x & F_x \\ C_x & D_x & G_x & H_x \\ E_y & F_y & A_y & B_y \\ G_y & H_y & C_y & D_y \end{pmatrix} \begin{pmatrix} n_1 p_1 \\ x_0 \\ n_1 q_1 \\ y_0 \end{pmatrix} \equiv \mathbf{T} \begin{pmatrix} n_1 p_1 \\ x_0 \\ n_1 q_1 \\ y_0 \end{pmatrix}. \quad (16)$$

The matrix elements $A_x \dots D_y$ are functions of the physical quantities defining the particular system.

We now want to complete the connection with ray-matrix optics by expressing L_N in Eq. (10) in terms of the matrix elements of **T**. The first step for doing this will be to rewrite the four simultaneous linear equations represented by matrix Eq. (16), treating $n_1 p_1$, $n_1 q_1$, $n_N p_N$, and $n_N q_N$ as unknowns. Then, solving the rewritten equations, we will obtain the four slopes in terms of x_0 , y_0 , x_N , y_N and the ray-matrix elements $A_x \dots D_y$. Finally, substitution into Eq. (10) will give L_N completely in terms of x_0 , y_0 , x_N , y_N and the ray-tracing matrix elements.

Thus, to solve for $n_1 p_1$, $n_1 q_1$, $n_N p_N$, and $n_N q_N$, we first put Eq. (16) in matrix form,

$$\mathbf{Ap} = \mathbf{Bx}, \quad (17a)$$

where

$$\mathbf{A} = - \begin{pmatrix} A_x & +1 & E_x & 0 \\ C_x & 0 & G_x & 0 \\ E_y & 0 & A_y & +1 \\ G_y & 0 & C_y & 0 \end{pmatrix}, \quad (17b)$$

$$\mathbf{B} = \begin{pmatrix} B_x & 0 & F_x & 0 \\ D_x & -1 & H_x & 0 \\ F_y & 0 & B_y & 0 \\ H_y & 0 & D_y & -1 \end{pmatrix}, \quad (17c)$$

$$\mathbf{p}^T = (-n_1 p_1, n_N p_N, -n_1 q_1, n_N q_N), \quad (17d)$$

and

$$\mathbf{x}^T = (x_0, x_N, y_0, y_N). \quad (17e)$$

The superscript T indicates transposed matrix or vector. Formally solving for **p** gives

$$\mathbf{p} = (\mathbf{A})^{-1} \mathbf{Bx}. \quad (17f)$$

The expression for L_N can then be written in terms of a

quadratic form by letting

$$L_N = L_0 + L_1, \quad (18a)$$

where

$$L_1 = \frac{1}{2} \mathbf{x}^T \mathbf{L} \mathbf{x} \quad (18b)$$

and

$$\mathbf{L} = (\mathbf{A})^{-1} \mathbf{B}. \quad (18c)$$

The matrix $\mathbf{L}$, obtained by inverting²⁰ $\mathbf{A}$ and forming the product $(\mathbf{A}^{-1})\mathbf{B}$, is

$$\mathbf{L} = \frac{1}{D} \begin{bmatrix} D_x C_y - G_x H_y & -C_y & H_x C_y - G_x D_y & G_x \\ L_{x1zN} & A_x C_y - E_x G_y & L_{y1zN} & C_x E_x - A_x G_x \\ C_x H_y - D_x G_y & G_y & C_x D_y - H_x G_y & -C_x \\ L_{x1y1} & A_y G_y - C_y E_y & L_{y1yN} & C_x A_y - G_x E_y \end{bmatrix}, \quad (19a)$$

where

$$L_{x1zN} = (C_x C_y - G_x G_y) B_x - (A_x C_y - E_x G_y) D_x + (A_x G_x - C_x E_x) H_y \quad (19b)$$

$$L_{x1yN} = (A_y G_y - C_y E_y) D_x - (C_x C_y - G_x G_y) F_y - (C_x A_y - G_x E_y) H_y \quad (19c)$$

$$L_{y1zN} = (C_x C_y - G_x G_y) F_x - (A_x C_y - E_x G_y) H_x + (A_x G_x - C_x E_x) D_y \quad (19d)$$

$$L_{y1yN} = -(E_y C_y - A_y G_y) H_x + (C_x C_y - G_x G_y) B_y - (C_x A_y - G_x E_y) D_y \quad (19e)$$

and D , the determinant of matrix $\mathbf{A}$, is given by

$$D = (C_x C_y - G_x G_y). \quad (19f)$$

Matrix $\mathbf{L}$ can be further simplified to put it in symmetric form. This is done in Appendix B, where it is shown that matrix elements L_{x1zN} , L_{x1yN} , L_{y1zN} , and L_{y1yN} can be simplified because of the form of ray-tracing matrices in Eqs. (14) and (15).

Equations (18), (19), and (11), along with Eqs. (14) and (15), constitute the results of this section, the derivation of the form of eikonal function in terms of the system ray matrix.

IV. DERIVATION OF AMPLITUDE FACTOR A_N

The most straightforward approach to the derivation of the amplitude factor A_N is to apply the conservation of energy law to the energy entering the input plane and leaving the output plane,

$$\begin{aligned} & \int \int |E_0(x_0, y_0)|^2 dx_0 dy_0 \\ &= \int \int |E_N(x_N, y_N)|^2 dx_N dy_N. \end{aligned} \quad (20)$$

The procedure is to insert into Eq. (20) the expression for E_N in Eq. (1) with the expression for $K(x_0, y_0, x_N, y_N)$ given in Eq. (2), and that for L_N given in Eq. (18). Making these substitutions, using the fact that L_N is

symmetric, and simplifying give

$$\begin{aligned} & \int \int |E_0(x_0, y_0)|^2 dx_0 dy_0 \\ &= |A_N|^2 \int \int \int \int \int dx_0 dy_0 dx'_0 dy'_0 dx_N dy_N \\ & \cdot E_0(x_0, y_0) E_0^*(x'_0, y'_0) \exp \left(\frac{ik}{2D} \{ L_{x0x0}(x_0^2 - x_0'^2) \right. \\ & + 2L_{x0y0}(x_0 y_0 - x_0' y_0') + L_{y0y0}(y_0^2 - y_0'^2) \\ & + 2x_N [-C_y(x_0 - x_0') + G_y(y_0 - y_0')] \\ & \left. + 2y_N [G_x(x_0 - x_0') - C_x(y_0 - y_0')] \} \right), \end{aligned} \quad (21a)$$

where

$$L_{x0x0} = D_x C_y - G_x H_y \quad (21b)$$

$$L_{x0y0} = C_x H_y - D_x G_y \quad (21c)$$

$$L_{y0y0} = C_x D_y - H_x G_y, \quad (21d)$$

and the asterisk indicates complex conjugate.

Performing the x_N and y_N integrations gives

$$\begin{aligned} & \int \int |E_0(x_0, y_0)|^2 dx_0 dy_0 = |A_N|^2 \int \int \int \int dx_0 dy_0 dx'_0 dy'_0 \\ & \cdot E(x_0, y_0) E^*(x'_0, y'_0) \exp \left\{ \frac{ik}{2D} [L_{x0x0}(x_0^2 - x_0'^2) \right. \\ & + 2L_{x0y0}(x_0 y_0 - x_0' y_0') + L_{y0y0}(y_0^2 - y_0'^2)] \} \\ & \times (2\pi)^2 \delta \left\{ \frac{k}{D} [-C_y(x_0 - x_0') + G_y(y_0 - y_0')] \right\} \\ & \times \delta \left\{ \frac{k}{D} [G_x(x_0 - x_0') - C_x(y_0 - y_0')] \right\}, \end{aligned} \quad (22)$$

where $\delta(x)$ is the Dirac delta function.

To perform the x_0 and y_0 integrations, we make the substitutions of variables

$$u = (k/D)[-C_y(x_0 - x_0') + G_y(y_0 - y_0')] \quad (23a)$$

$$v = (k/D)[G_x(x_0 - x_0') - C_x(y_0 - y_0')] \quad (23b)$$

$$dx_0 dy_0 = (D/k^2) du dv. \quad (23c)$$

Performing the x_0 and y_0 integrations and canceling the integrals give

$$A_N = K'(k/2\pi)(C_x C_y - G_x G_y)^{-1}, \quad (24)$$

where K' is a complex constant of unit magnitude.

The constant K' can be evaluated most simply by comparing the final integral expression, Eq. (1), written for the case of a single translation to the right of the input plane, with an expression for the same integral described by Huygens-Fresnel-Kirchhoff diffraction theory. Thus we write Eq. (1) with $K(x_0, y_0; x_N, y_N)$ from Eqs. (2), (18), (19), and (24) and with $A_x = A_y = D_x = D_y = 1$, $C_x = C_y = d/n_1$, and $E_x \cdots H_y = 0$. Comparing with the diffraction integral written in the sagittal approximation gives

$$K' = -i. \quad (25) \quad \text{and}$$

$$L = (C_x C_y - G_x G_y)^{-1} \begin{bmatrix} C_y D_x - G_x H_y & -C_y & H_x C_y - G_x D_y & G_x \\ -C_y & A_x C_y - E_x G_y & G_y & C_x E_x - A_x G_x \\ H_x C_y - D_y G_x & G_y & C_x D_y - H_x G_y & -C_x \\ G_x & C_x E_x - A_x G_x & -C_x & C_x A_y - G_x E_y \end{bmatrix}. \quad (27d)$$

Thus $K(x_0, y_0; x_N, y_N)$ is written in terms of elements from the system ray matrix T , Eq. (16), as desired.

Equations (26) and (27) are the general result of this paper, an expression for the diffraction integral for an optical system possibly employing nonsymmetric

Finally, the complete result can be written by combining Eqs. (1), (2), (11), (17)–(19), (24), and (25) with the simplifications of Appendix B. We take the phase to be zero at x_0, y_0, z_0 . Then

$$K(x_0, y_0; x_N, y_N) = A_N \exp(ikL_N), \quad (26a)$$

where $L_N = L_0 + L_1$. Further,

$$E_N(x_N, y_N) = -\frac{ik}{2\pi} (C_x C_y - G_x G_y)^{-1} \exp(ikL_0) \times \int \int E_N(x_0, y_0) \exp[ikL_1(x_0, y_0; x_N, y_N)] dx_0 dy_0, \quad (26b)$$

where

$$L_0 = \sum_{i=1}^N n_i(z_{0i} - z_{0i-1}), \quad (27a)$$

$$L_1 = \frac{1}{2} \mathbf{x}^T \mathbf{L} \mathbf{x}, \quad (27b)$$

$$\mathbf{x}^T = (x_0, x_N, y_0, y_N), \quad (27c)$$

elements, written in terms of the elements of the ray-transfer matrix for that system.

Equation (26b) reduces to some useful and some easily recognized expressions. The form perhaps most useful deals with systems employing spherical optics. For such a case, Eq. (26) simplifies to

$$E_N(x_N, y_N) = -\frac{ik \exp(ikL_0)}{2\pi C} \int \int E_0(x_0, y_0) \exp \frac{ik}{2C} [(x_0^2 + y_0^2)D - 2(x_0 x_N + y_0 y_N) + A(x_N^2 + y_N^2)] dx_0 dy_0, \quad (28)$$

where $A = A_x = A_y$, $C = C_x = C_y$, $D = D_x = D_y$.

For a single thin lens of focal length, f , separated from input and output planes by distances d_1 and d_2 , respectively, recourse to ray matrices^{3,4} gives

$$A = \frac{f-d_1}{f}, \quad C = \frac{f^2 - (f-d_1)(f-d_2)}{f}, \quad D = \frac{f-d_2}{f},$$

and the well-known expression,

$$E_2(x_2, y_2) = -\frac{ik}{2\pi} \frac{f e^{ikL_0}}{[f^2 - (f-d_1)(f-d_2)]} \int \int E_0(x_0, y_0) \times \exp \left(\frac{ik[(x_0^2 + y_0^2)(f-d_2) - 2f(x_0 x_2 + y_0 y_2) + (f-d_1)(x_2^2 + y_2^2)]}{2[f^2 - (f-d_1)(f-d_2)]} \right) dx_0 dy_0. \quad (29)$$

If $d_1 = d_2$ the well-known Fourier-transform expression results, and if $f \rightarrow \infty$, the Fresnel diffraction integral

results. If the input and output planes are conjugate planes, then the constant C in Eq. (28) equals zero,

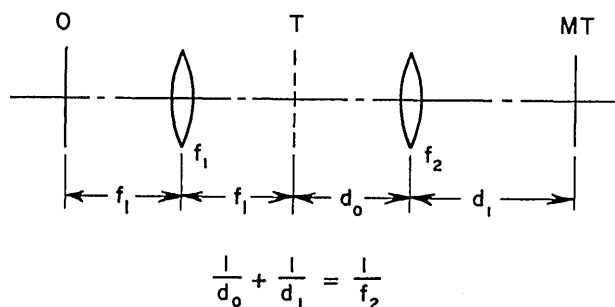


FIG. 2. Schematic illustration of an optical system magnifying a Fourier-transform plane. O is the object plane, T is the transform plane, and MT is the magnified transform plane.

making the integral undefined. For that case a limiting process can be used to show, at least formally, that the fields E_0 and E_N are proportional.

V. APPLICATIONS

As a first simple case, consider a symmetric optical system designed to perform an optical Fourier transform and then to magnify the transform plane. The arrangement is shown in Fig. 2. The ray matrix for the complete system is determined by

$$\begin{aligned} \begin{bmatrix} n_N p_N \\ x_N \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ \frac{d_1}{n_0} & 1 \end{bmatrix} \begin{bmatrix} 1 & \frac{-n_0}{f_2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{d_0}{n_0} & 1 \end{bmatrix} \\ &\times \begin{bmatrix} 1 & 0 \\ \frac{f_1}{n_0} & 1 \end{bmatrix} \begin{bmatrix} 1 & \frac{-n_0}{f_1} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{f_1}{n_0} & 1 \end{bmatrix} \begin{bmatrix} n_1 p_1 \\ x_0 \end{bmatrix} \\ &= \begin{bmatrix} -\frac{f_1}{f_2} & \frac{n_0}{f_1} \left(1 - \frac{d_0}{f_2}\right) \\ \frac{f_1}{n_0} \left(1 - \frac{d_1}{f_2}\right) & 0 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}. \end{aligned} \quad (30)$$

We note that, because $A = -f_1/f_2 \neq 0$, there is a term in $(x_N^2 + y_N^2)$ in the exponent of the kernel in Eq. (28) that destroys the exact Fourier-transform relationship. This quadratic phase factor indicates that there will be a curved wave front on which the Fourier transform is superimposed.

In order to ascertain the position of input and output planes that give an exact Fourier-transform relationship, we can reconsider the problem, treating the distances from input plane to first lens and from output plane to last lens as unknowns and retaining the lens separation. The solution is straightforward, with the

result that there is an exact Fourier transform with no quadratic phase terms between the focal planes of the two-lens pair.

As a second application, the formalism allows for derivation and extension of the $ABCD$ law^{9,10} of Kogelnik for gaussian-beam propagation in an optical system, which he derived by physical reasoning.¹⁰ To do this, consider, for simplicity, a symmetrical optical system. Let the input field be

$$E_0(x_0, y_0) = -\frac{2^{1/2}}{w_0} H_n\left(\frac{x_0}{w_0}\right) H_m\left(\frac{y_0}{w_0}\right) \exp\left(\frac{ik}{2q_0}(x_0^2 + y_0^2)\right), \quad (31)$$

where q_0 is related to the spot size w_0 and wave-front radius R_0 by $q_0^{-1} = R_0^{-1} + (2i/kw_0^2)$. After Eq. (31) is inserted into Eq. (28), performing the integration (taking $n_1 = n_N = 1$) yields

$$\begin{aligned} E_N(x_N, y_N) &= \exp(ikL_0) -\frac{2^{1/2}}{w_N} H_n\left(\frac{x_N}{w_N}\right) H_m\left(\frac{y_N}{w_N}\right) \\ &\times \exp\left[\frac{ik}{2q_N}(x_N^2 + y_N^2)\right] \cdot \exp[-i\alpha_0(n+m+1)], \end{aligned} \quad (32a)$$

where

$$\frac{1}{q_N} = (A + Bq_0)/C + Dq_0 \quad (32b)$$

$$w_N = w_0 C [(D/C + 1/R_0)^2 + (2/kw_0^2)^2]^{1/2} \quad (32c)$$

Equation (32b) is the $ABCD$ law²¹ derived previously for zero-order gaussian modes. Equations (32a) and (32c) represent the extension to higher-order gaussian-hermite modes.

Next, consider the optical resonator formed by putting flat mirrors on the input and output planes of an optical system. For simplicity, consider only sym-

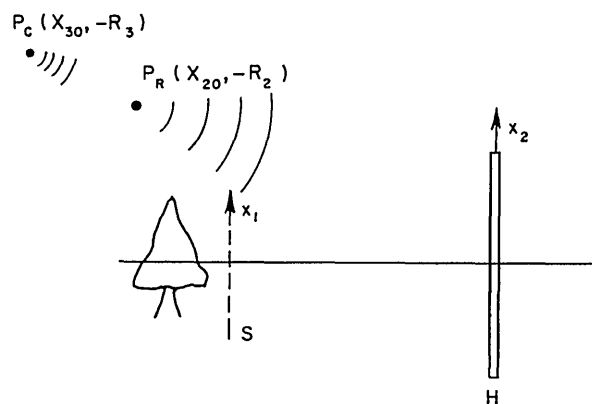


FIG. 3. Schematic illustration of the holographic system. S and H are, respectively, the signal and hologram planes. P_r is the reference point source, and P_c is the reconstruction point source.

metric lens elements. The situation has been considered previously.¹ Following Ref. 1 we put $\rho_{11}=D/2C$, $\rho_{nn}=A/2C$, $\rho_{1n}=1/C$. The spot sizes w_0 and w_N on the input and output plane mirrors, respectively, are then

$$w_0 = \left[-(4/k^2)(AC/BD) \right]^{\frac{1}{2}}$$

and

$$w_N = \left[-(4/k^2)(CD/AB) \right]^{\frac{1}{2}}.$$

Further, the resonant frequencies are $\nu_{q,m,n} = c/2L_0 \times [q + (m+n+1)(\alpha/\pi)]$, where $\alpha = \cos^{-1}(AD)$. The stability conditions²² are $AD \leq 1$ and $A/D \geq 0$.

As an example of a somewhat different application consider a description of a reconstructed holographic image. The situations for making and reconstructing the hologram are shown in Fig. 3. For simplicity, only one transverse dimension is shown; the behavior in the other transverse dimension is identical. The signal is represented by $E(x_1, -d)$ across a plane perpendicular to the axis at $z = -d$. The hologram is in the plane $(x_2, 0)$. The reference and reconstruction beams are spherical waves centered at $(x_{20}, -R_2)$ and $(x_{30}, -R_3)$, respectively. It will also be assumed that the reference and reconstruction beams are in the same direction, so that $x_{20}/R_2 = x_{30}/R_3$. The reconstructed virtual-image fields just to the right of the plane at $(x_2, 0)$ are (neglecting a complex constant)

$$E_R(x_2, 0) = \exp \left[ik(d - R_2 - R_3) + \frac{ik(x_2 - x_{30})^2}{2R_3} - \frac{ik(x_2 - x_{20})^2}{2R_2} \right] \int E(x_1) \times \exp \left(\frac{ik(x_2 - x_1)^2}{2d} \right) dx_1 \quad (33a)$$

$$= \exp \left[ik(d - R_2 + R_3) + ik \left(\frac{x_{30}^2}{R_3} - \frac{x_{20}^2}{R_2} \right) + ik(z) \right] \cdot \int E(x_1) \exp \left[\frac{ik}{2} \left(\frac{x_1^2}{d} - \frac{2x_1x_2}{d} \right) + x_2^2 \left(\frac{1}{R_3} - \frac{1}{R_2} + \frac{1}{d} \right) \right] dx_1. \quad (33b)$$

The integral expression can be used to identify the geometrical-optics matrix elements for the hologram. Equating the coefficients of x_1^2 , $2x_1x_2$, and x_2^2 with D/C , $1/C$, and A/C , respectively, gives the equivalent

TABLE I. Key to evaluation of terms t_{ij} in Eqs. (36).

First subscript	Rows	Second subscript	Columns
2	23	1	13
3	12	2	23
4	14	3	12
5	24	4	14
		5	24
$t_{33} = +(A_x D_x - B_x C_x) - (E_x H_x - F_x G_x)$			

ray matrix

$$\mathbf{T} = \begin{pmatrix} \left(\frac{1}{R_3} - \frac{1}{R_2} + \frac{1}{d} \right) & \left(\frac{1}{R_3} - \frac{1}{R_2} \right) \\ d & 1 \end{pmatrix} = \begin{bmatrix} 1 & \left(\frac{1}{R_3} - \frac{1}{R_2} \right) \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ d & 1 \end{bmatrix}. \quad (34)$$

The factored form of $\mathbf{T}$ indicates that the transformations equivalent to the hologram are the translation through distance d and refraction by a lens of focal length $f = -(1/R_3 - 1/R_2)^{-1}$. For the real image, the equivalent-lens focal length can be shown to be $+(1/R_3 - 1/R_2)^{-1}$. These equivalents were indicated by Lukosz.^{23,24}

For an application to a problem with reduced symmetry, we consider the general case of a rotated, astigmatic, gaussian beam incident on an asymmetrical optical system, as considered by Suematsu *et al.*²⁵ Let the form of the beam across the input plane be

$$E_0(x_0, y_0) = E_0 \exp[i(k/2)(P_x x_0^2 - 2P_{xy} x_0 y_0 + P_y y_0^2)], \quad (35)$$

where P_x , P_{xy} , and P_y are complex constants associated with the spot size and wave-front shape. We insert Eq. (35) into Eq. (28), and perform the resulting and somewhat tedious integration (assuming a stable optical system).^{13,26} Simplifying yields

$$E_N(x_N, y_N) = (D_x D_y - H_x H_y)^{-\frac{1}{2}} \times \exp \left[ikL_0 + \frac{ik}{2} (P'_x x_N^2 - 2P'_{xy} x_N y_N + P'_y y_N^2) \right], \quad (36a)$$

where the transformation equations for the P 's are given by

$$P'_x = \frac{t_{41}(P_x P_y - P_{xy}^2) + t_{42} P_y - 2t_{43} P_{xy} + t_{44} P_x + t_{45}}{P} \quad (36b)$$

$$P'_{xy} = \frac{t_{31}(P_x P_y - P_{xy}^2) + t_{32} P_y - t_{33} P_{xy} + t_{34} P_x + t_{35}}{P} \quad (36c)$$

$$P_y' = \frac{t_{21}(P_x P_y - P_{xy}^2) + t_{22}P_y - 2t_{23}P_{xy} + t_{24}P_x + t_{25}}{P} \quad (36d)$$

$$P = t_{51}(P_x P_y - P_{xy}^2) + t_{52}P_y - 2t_{53}P_{xy} + t_{54}P_x + t_{55}. \quad (36e)$$

The coefficients t_{ij} are second-order determinants composed of elements of the ray matrix $\mathbf{T}$, given in Eq. (16). The determinants are formed from elements at the intersection of two rows and two columns of that matrix. The appropriate rows and columns for each t_{ij} are indicated in Table I. For example, t_{21} is formed by the second-order determinant from the second and third rows and first and third columns of the ray matrix. Thus $t_{21} = C_x A_y - G_x E_y$. The amplitude factor in Eq. (36a) is new. The rest of the results were given by Suematsu and Fukinuki,¹³ derived presumably on physical grounds.

VI. SUMMARY AND CONCLUSIONS

In this paper, an expression is described relating fields across the input and output planes of an optical system. The kernel of the integral is a function of the elements of the ray-tracing matrix. The situation considered is sufficiently general to include optical systems without two planes of symmetry, i.e., with skewed elliptical lenses or mirrors; but is limited to paraxial-ray optics. The expression unifies and extends the results of several authors in the areas of gaussian-beam propagation through optical systems, optical data processing, and holography.

APPENDIX A

In this appendix, we present a few steps that simplify the process of optimizing L' in Eq. (6). The first step of the optimization involves inserting z_i in Eq. (6) into the first two terms in the summation in Eq. (5), and rearranging terms within the summation, so as to rewrite it in the form

$$L_N' = \sum_{i=1}^N n_i(z_{0i} - z_{0i-1}) + \frac{n_1(x_0^2 - x_0 x_1)}{2(z_{01} - z_{00})} + \frac{n_1(y_0^2 - y_0 y_1)}{2(z_{01} - z_{00})} + \frac{n_N(x_N^2 - x_N x_{N-1})}{2(z_{0N} - z_{0N-1})} + \frac{n_N(y_N^2 - y_N y_{N-1})}{2(z_{0N} - z_{0N-1})} \\ + \sum_{i=1}^N \left\{ \frac{x_i}{2} \left[-\frac{n_i x_{i-1}}{(z_{0i} - z_{0i-1})} + x_i \left[\frac{n_i}{(z_{0i} - z_{0i-1})} + \frac{n_{i+1}}{(z_{0i+1} - z_{0i})} - 2c_i(n_{i+1} - n_i) \right] - \frac{n_{i+1} x_{i+1}}{(z_{0i+1} - z_{0i})} - d_i y_i(n_{i+1} - n_i) \right] \right. \\ \left. + \frac{y_i}{2} \left[-\frac{n_i y_{i-1}}{(z_{0i} - z_{0i-1})} + y_i \left[\frac{n_i}{(z_{0i} - z_{0i-1})} + \frac{n_{i+1}}{(z_{0i+1} - z_{0i})} - 2e_i(n_{i+1} - n_i) \right] - \frac{n_{i+1} y_{i+1}}{(z_{0i+1} - z_{0i})} - d_i x_i(n_{i+1} - n_i) \right] \right\}. \quad (A1)$$

To optimize L_N' , differentiate the original unarranged expression with respect to the x_i or y_i to give equations for the optimized values of x_i and y_i (call them x_{0i} and y_{0i}). The equations are

$$\frac{\partial L_N'}{\partial x_i} = 0 = -\frac{n_i x_{0i-1}}{(z_{0i} - z_{0i-1})} + x_{0i} \left[\frac{n_i}{(z_{0i} - z_{0i-1})} + \frac{n_{i+1}}{(z_{0i+1} - z_{0i})} - 2c_i(n_{i+1} - n_i) \right] - \frac{n_{i+1} x_{0i+1}}{(z_{0i+1} - z_{0i})} - y_{0i} d_i(n_{i+1} - n_i) \quad (A2)$$

and

$$\frac{\partial L_N'}{\partial y_i} = 0 = -\frac{n_i y_{0i-1}}{(z_{0i} - z_{0i-1})} + y_{0i} \left[\frac{n_i}{(z_{0i} - z_{0i-1})} + \frac{n_{i+1}}{(z_{0i+1} - z_{0i})} - 2e_i(n_{i+1} - n_i) \right] - \frac{n_{i+1} y_{0i+1}}{(z_{0i+1} - z_{0i})} - x_{0i} d_i(n_{i+1} - n_i). \quad (A3)$$

Equation (A1) has been partitioned so that the second summation term in the optimized value of L_N' is identically zero. This follows from comparison of the terms in the summation with terms in Eqs. (A2) and (A3). We will designate the optimized value of L_N' as L_N .

Equation (8) in the text results.

APPENDIX B

In this appendix, we develop the identities necessary to show that the matrix $\mathbf{L}$ is symmetric. These identities belong to sets, all proven in the same manner. The sets

will be listed for completeness. We consider first the identity $C_x H_y - D_x G_y = H_x C_y - G_x D_y$ required to show that $\mathbf{L}$ is symmetric. This is one of the following set of four identities:

$$A_x F_y - B_x E_y = A_y F_x - B_y E_x, \quad (B1)$$

$$A_x H_y - B_x G_y = C_y F_x - D_y E_x, \quad (B2)$$

$$C_x F_y - D_x E_y = A_y H_x - B_y G_x, \quad (B3)$$

$$C_x H_y - D_x G_y = C_y H_x - D_y G_x. \quad (B4)$$

The identities are all proved by induction. To demon-

strate the proofs, note that they all hold for a single translation matrix of the type shown in Eq. (15), call it $\mathbf{t}$, or, for a single refraction matrix of the type shown in Eq. (14), denoted by $\mathbf{r}$. Further, if the identities are assumed to hold for a given general system matrix $\mathbf{T}$ of the type of Eq. (16), they can also be shown, quite simply, to hold for the matrix products $\mathbf{Tt}$ and $\mathbf{Tr}$, QED.

The following identities including the required one, $A_y G_y - C_y E_y = A_x G_x - C_x E_x$, can be proved in the same fashion, except that the matrix $\mathbf{T}$ is multiplied from the left rather than from the right by the matrices $\mathbf{t}$ and $\mathbf{r}$:

$$A_x G_x - C_x E_x = A_y G_y - C_y E_y, \quad (\text{B5})$$

$$A_x H_x - C_x F_x = B_y G_y - D_y E_y, \quad (\text{B6})$$

$$B_x G_x - D_x E_x = A_y H_y - C_y F_y, \quad (\text{B7})$$

$$B_x H_x - D_x F_x = B_y H_y - D_y F_y. \quad (\text{B8})$$

The following identities can be proved in similar fashion, by following the induction procedure used for Eqs. (B1)–(B4) and using, where needed, Eqs. (B5)–(B8):

$$A_x D_x - B_x C_x + E_x H_x - F_x G_x = 1, \quad (\text{B9})$$

$$A_y D_y - B_y C_y + E_y H_y - F_y G_y = 1, \quad (\text{B10})$$

$$A_x D_x - B_x C_x + E_y H_y - F_y G_y = 1, \quad (\text{B11})$$

$$A_y D_y - B_y C_y + E_x H_x - F_x G_x = 1. \quad (\text{B12})$$

Equations (B9)–(B12) are the extensions to asymmetric lens systems of the identity $AD - BC = 1$, which is valid for symmetric lens systems.

Finally, the remaining identities, $L_{x1xN} = -C_y$, $L_{x1yN} = G_x$, $L_{y1xN} = G_y$, and $L_{y1yN} = -C_x$, necessary to show that $\mathbf{L}$ is symmetric, can be demonstrated by use of Eqs. (B1)–(B12). For example, the identity $L_{x1xN} = -C_y$ can be easily verified by use of identities B2, B4, and B9.

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- ²⁰ F. B. Hildebrandt, *Methods of Applied Mathematics* (Prentice-Hall, New York, 1952), Sec. 1.6.
- ²¹ To show complete agreement, it is necessary to change notation by putting A to C , B to D , C to A , and D to B to make our ray matrix [Eq. (16)] consistent with Kogelnik's Eq. (9).
- ²² This stability condition can be brought into agreement with the stability condition $0 < AD < 1$ predicted on the basis of lens-waveguide theory. To show the equivalence, it is necessary to form the equivalent lens-waveguide-unit lens system from the unit mentioned here, followed by this unit, inverted end for end.
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- ²⁴ For more complicated cases in which L_N possesses a linear term in x_1 or in x_2 it is necessary to use Hamilton's equations. $n_i p_i = (-1)^i \partial L / \partial x_i$, where $i = 1, 2$, to derive the complete ray-matrix equations.
- ²⁵ The case considered in Ref. 13 is slightly different, in that they start with the fields across the output plane and derive the input fields. They also incorporate the factor of (ik) in the exponent of Eq. (35) into the P 's without significant change of results.
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